Answer **one** question from this section in the spaces provided.

8 **(a)** Point source P, consisting of light with wavelength 630 nm, passes through a narrow slit and is incident on a screen at a distance of 2.4 m from the slit. Fig. 8.1 below shows the variation of intensity I of the light on the screen with distance x along the screen.

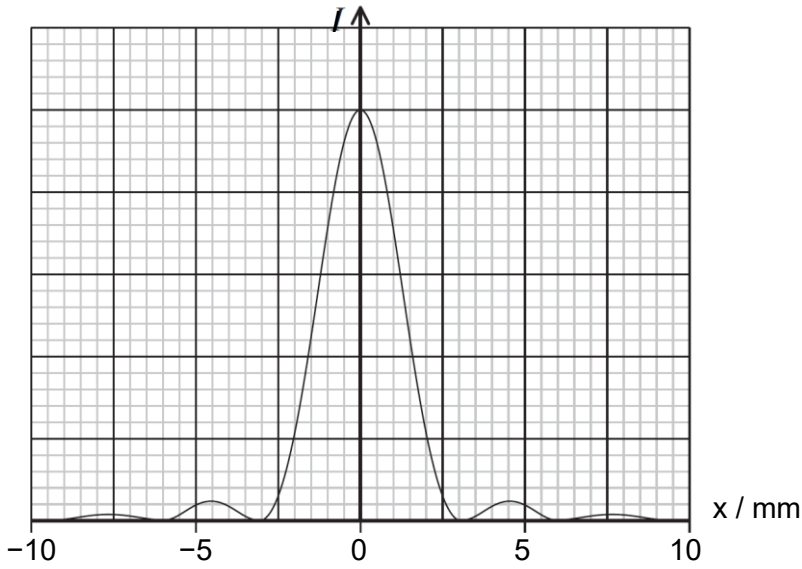

Fig. 8.1

(i) Use Fig. 8.1 to determine the width of the slit.

width = ... mm [2]

(ii) State the effect on the pattern on the screen if each of the following changes is made separately:

1. the width of the single slit is reduced,

.....

.....

..... [2]

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2. the red source is replaced with another source of violet light.

.....

.....

..... [2]

(b) Another point source Q, emitting light of the same wavelength as P, is placed 5.0 mm from P, and the two sources are arranged as shown in Fig. 8.2. The slit is now replaced with another slit of width *b*.

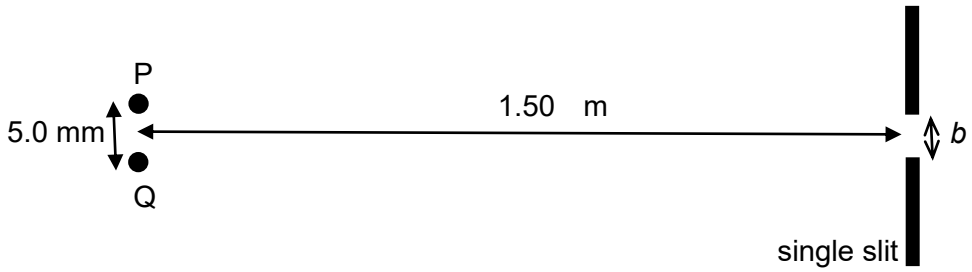

Fig. 8.2

(i) State the condition for the image of P and the image of Q formed on the screen to be just resolved.

.....

..... [1]

(ii) Sketch, on Fig. 8.1, the graph for point source Q when the condition in **(b)(i)** is met. [1]

(iii) Calculate the minimum width *b* of the slit for the two images to be just resolved.

b = ... mm [3]

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(c) Two microwave transmitters, A and B, are separated by a distance of 0.80 m. They each transmit a microwave of frequency 3.00 GHz.

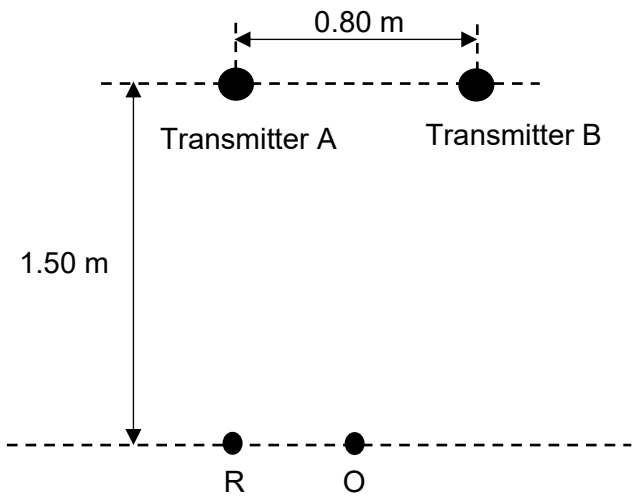

Fig. 8.3

A microwave receiver is moved along a line parallel to, and 1.50 m from, the line joining the two transmitters, as shown in Fig. 8.3. Point O lies on the line where it is equidistant from both transmitters, while point R is at a distance 1.50 m from transmitter A. As the receiver moves from O to R, a minimum signal is detected at R.

(i) Determine the path difference between the two waves arriving at R.

path difference = …… m [2]

(ii) Show that the two waves are emitted in antiphase at the transmitters.

[4]

Fig. 8.4 below shows the displacement-time graph of the waves arriving at R when only transmitter A is switched on and when both transmitters A and B are switched on.

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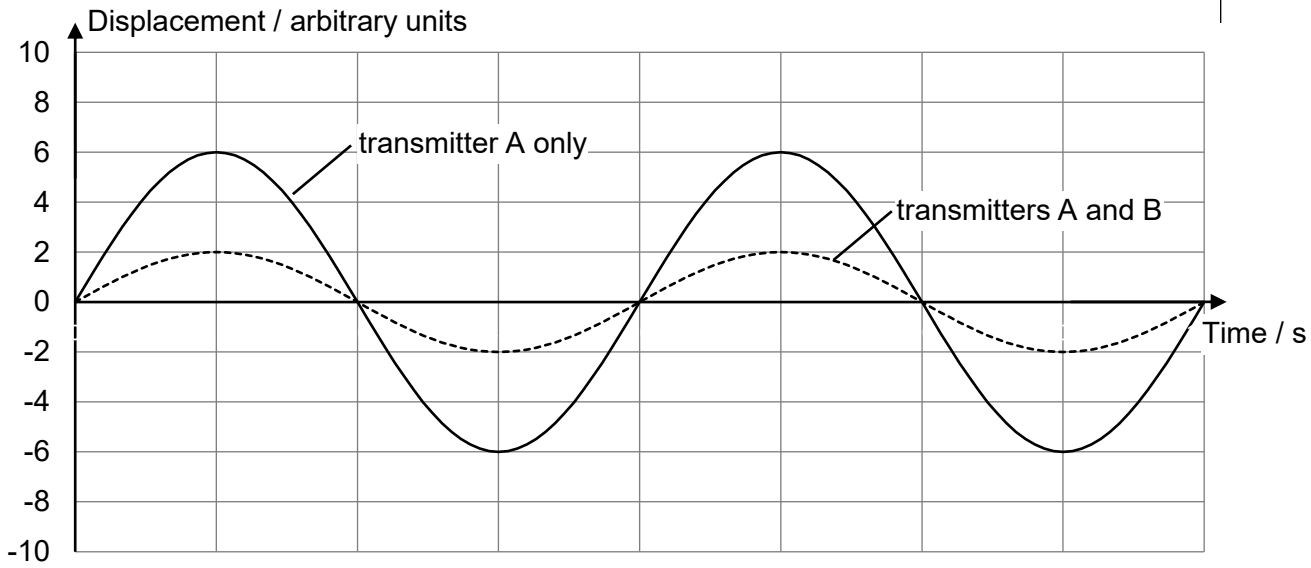

Fig. 8.4

(iii) Sketch, on Fig. 8.4, the displacement-time graph of the waves arriving at R when only transmitter B is switched on. [1]

(iv) Determine the ratio, at point R, of $\frac{\text{intensity of microwave due to transmitter A}}{\text{intensity of microwave due to transmitter B}}$.

ratio = ... [2]